

Cardiac Arrhythmias: Approach to a Case of Palpitation and/or Syncope

Dr. Muhammad Reihan, Internal Medicine | Source: Approach to a case of palpitation and/or syncope – Cardiac Arrhythmia (86 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 Bridge from earlier lectures:

- **CAD: right coronary (inferior) MI → bradycardia/AV block.** VT/VF is the early killer after MI.
- **Valvular heart disease: mitral stenosis → AF** (LA enlargement) → stroke. "Valvular AF" is an automatic anticoagulation indication.
- **Thyroid 1: thyrotoxicosis → sinus tachycardia/AF.**
- **Hypo/hyperparathyroidism: hypocalcaemia → long QT → torsades;** hypercalcaemia → short QT.
- **COPD/PE lectures: MAT** in COPD exacerbation; sinus tachycardia is the commonest ECG finding in PE.
- **Pericardial: electrical alternans** in tamponade.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Clinical features and ECG characteristics of the major brady- and tachyarrhythmias	ECG strip → name
Knowledge	★ Components of CHA₂DS₂-VASc and its role in AF anticoagulation	Calculate the score → OAC or not
Knowledge	Mechanisms of AVNRT, AVRT, AF, VT	Dual pathway; accessory pathway
Skill	★ Classify tachyarrhythmia by QRS width and regularity	The 4 groups
Skill	★ Rate vs rhythm control + anticoagulation plan in AF	Drug choice; > 48 h rule
Skill	★ Emergency plan for the unstable tachyarrhythmia	Synchronised DC shock

Lecture flow: Opening case → Definitions → Causes (brady, tachy) → Clinical picture → Classification → **Bradyarrhythmias** (sinus, AV blocks) → Bradycardia management → **Tachyarrhythmias** (SVT types, VT types, ECG gallery) → Tachycardia management (stable/unstable; 4 groups) → Narrow regular (sinus tach, AVNRT, AVRT) → Narrow irregular (**AF**) → Wide regular (**VT**, TdP) → Wide irregular (VF, **AF + WPW**) → Non-pharmacological therapy (vagal, cardioversion, ablation, pacing) → Drug classes → Summary algorithm.

2. Main Content (Lecture Order)

2.0 Opening Case (Slides 5-6)

- **Patient:** 16-year-old boy with palpitations and light-headedness for 16 h after **binge drinking** ("holiday heart" *[added]*). He was sedated and **electrically cardioverted**.
- **ECG after cardioversion:** HR 80, ★ **short PR + wide QRS with a slurred upstroke** (= **delta wave** → **WPW**). Electrolytes and creatinine normal.
- **12 h later:** palpitations again, **HR 220**, BP **112/84** (stable), SpO₂ 98%. ECG: **regular, short PR, wide monomorphic QRS**.

→ Next step? (A) synchronised cardioversion · (B) MgSO₄ · (C) verapamil · (D) atenolol · (E) **procainamide** · (F) adenosine. (*Answered in §7, Q1.*)

2.1 Definitions (Slide 7)

Arrhythmias = accelerated, slowed or irregular heart rates caused by abnormal myocardial electrical impulses.

Term	Rate
Bradyarrhythmia	< 60/min
Tachyarrhythmia	> 100/min

2.2 Causes (Slides 8-9)

	Bradyarrhythmia	Tachyarrhythmia
Physiological	Athletes	Physiological response in healthy people
Cardiac	★ Sick sinus syndrome ; IHD, valvular disease; ↑ vagal tone ; SA and AV conduction disease; ★ right coronary (inferior) infarction	Structural disease (VHD, myocarditis, MI, CHF); sympathetic activation or vagal withdrawal; ectopic foci (single/multiple); accessory pathways and macro-re-entry ; ★ channelopathies: congenital long QT, Brugada, CPVT
Extracardiac	★ Drugs: digoxin, β-blockers, CCBs <i>[added]</i> also hypothyroidism, hyperkalaemia, hypothermia, ↑ ICP)	Electrolyte imbalance; COPD exacerbation, PE, thyrotoxicosis ; drugs: ★ digitalis toxicity, theophylline

Clinical picture (slide 10):

- Often **asymptomatic**.
- When symptomatic: - **palpitations** - **light-headedness** - **dyspnoea** - ★ **syncope**

2.3 Classification (Slides 11-12, 24-25)

- **Supraventricular:** origin in the **SA node, atrial myocardium or AV node** (usually narrow QRS).
- **Ventricular:** origin **below the AV node** (★ **wide QRS**).

	Bradyarrhythmias	Supraventricular tachy	Ventricular tachy
Atrial / sinus	Respiratory sinus arrhythmia · sinus bradycardia · sinus pause/arrest · tachy-brady syndrome	SV premature beats · sinus tachycardia · atrial flutter · AF · atrial tachycardia (~5%) · MAT	—

	Bradyarrhythmias	Supraventricular tachy	Ventricular tachy
AV node	AV block: 1st · 2nd (Mobitz I / Wenckebach, Mobitz II) · 3rd	AVRT (orthodromic = narrow; antidromic = wide with delta waves) · AVNRT · junctional tachycardia	—
Ventricle	—	—	Monomorphic VT (most common) · polymorphic VT · torsades de pointes · VF

🔄 **Recap: Framework.** Brady < 60, tachy > 100. Causes: physiological (athletes), cardiac (SSS, IHD, inferior MI, accessory pathways, channelopathies), extracardiac (drugs: digoxin, β -blockers, CCB; electrolytes; thyroid; PE; COPD). Symptoms: palpitations, light-headedness, dyspnoea, syncope. SVT starts above or at the AV node (narrow); VT starts below (wide).

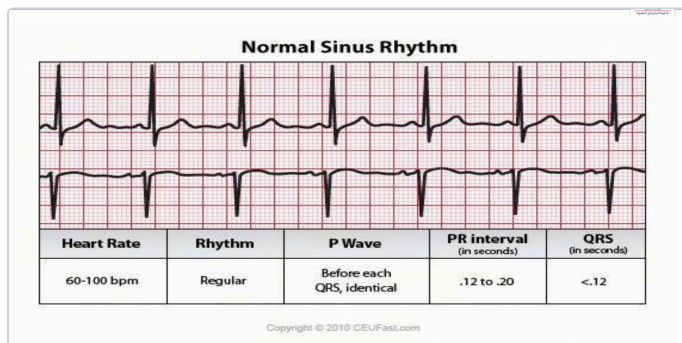
🔗 **Link:** The slow rhythms come first, from the SA node down to complete heart block.

▶ **Up next: Bradyarrhythmias.**

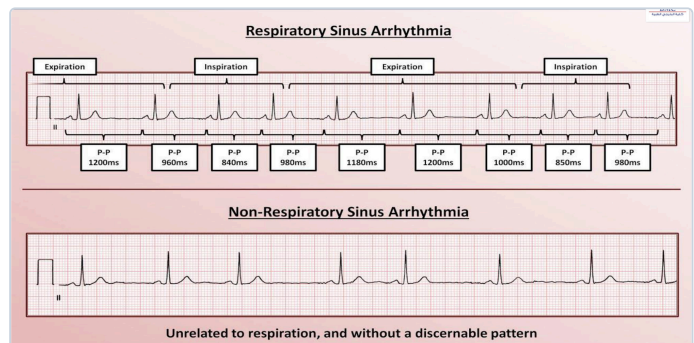
2.4 Bradyarrhythmia ECG Gallery (Slides 13-21)

Normal sinus rhythm (slide 13):

- HR **60-100**, regular
- A **P** before each **QRS**, all identical
- **PR 0.12-0.20 s**
- **QRS < 0.12 s**



Slide 13: normal sinus rhythm criteria

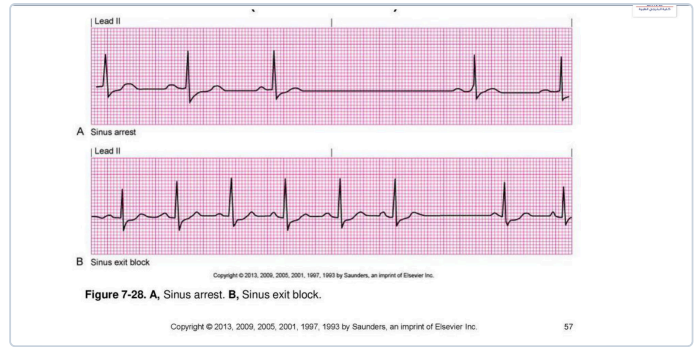


Slide 15: respiratory vs non-respiratory sinus arrhythmia

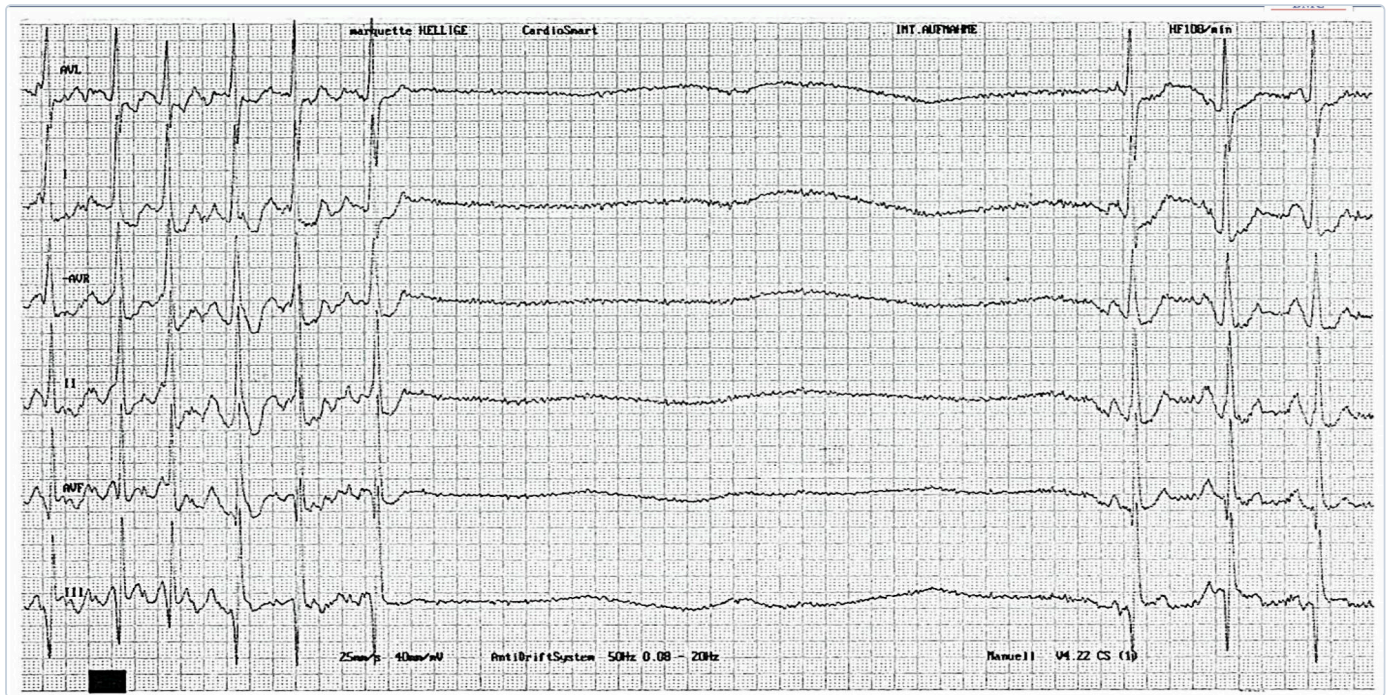
Rhythm	ECG key [added explanation]	Significance
Respiratory sinus arrhythmia	P-P shortens on inspiration , lengthens on expiration	Normal (young people); non-respiratory variation has no respiratory pattern
Sinus bradycardia	Normal P-QRS, rate < 60 (slide example HR 37)	Athletes, drugs, inferior MI, hypothyroid
Sinus pause / arrest	Sudden absent P-QRS; sinus exit block = pause an exact multiple of the P-P interval	Sick sinus syndrome
★ Tachy-brady syndrome	Runs of AF/flutter/SVT followed by long pauses on stopping	Form of sick sinus syndrome → pacemaker + rate control



Slide 16: sinus bradycardia (HR 37)



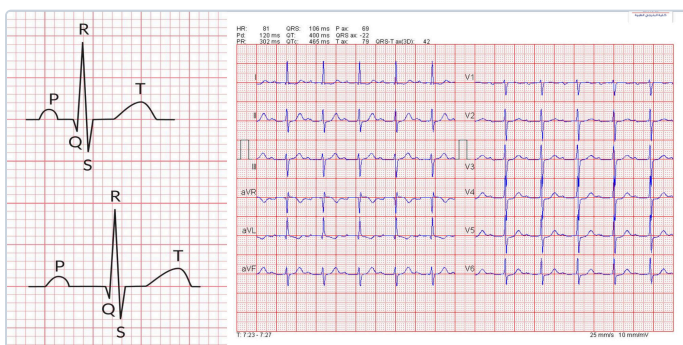
Slide 17: sinus arrest vs sinus exit block



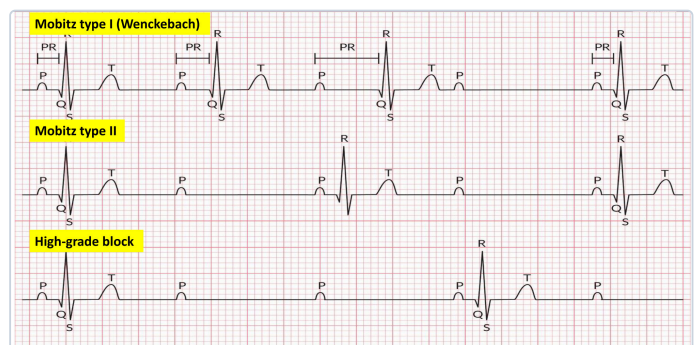
Slide 18: tachycardia-bradycardia syndrome

AV blocks (slides 19-21) ★

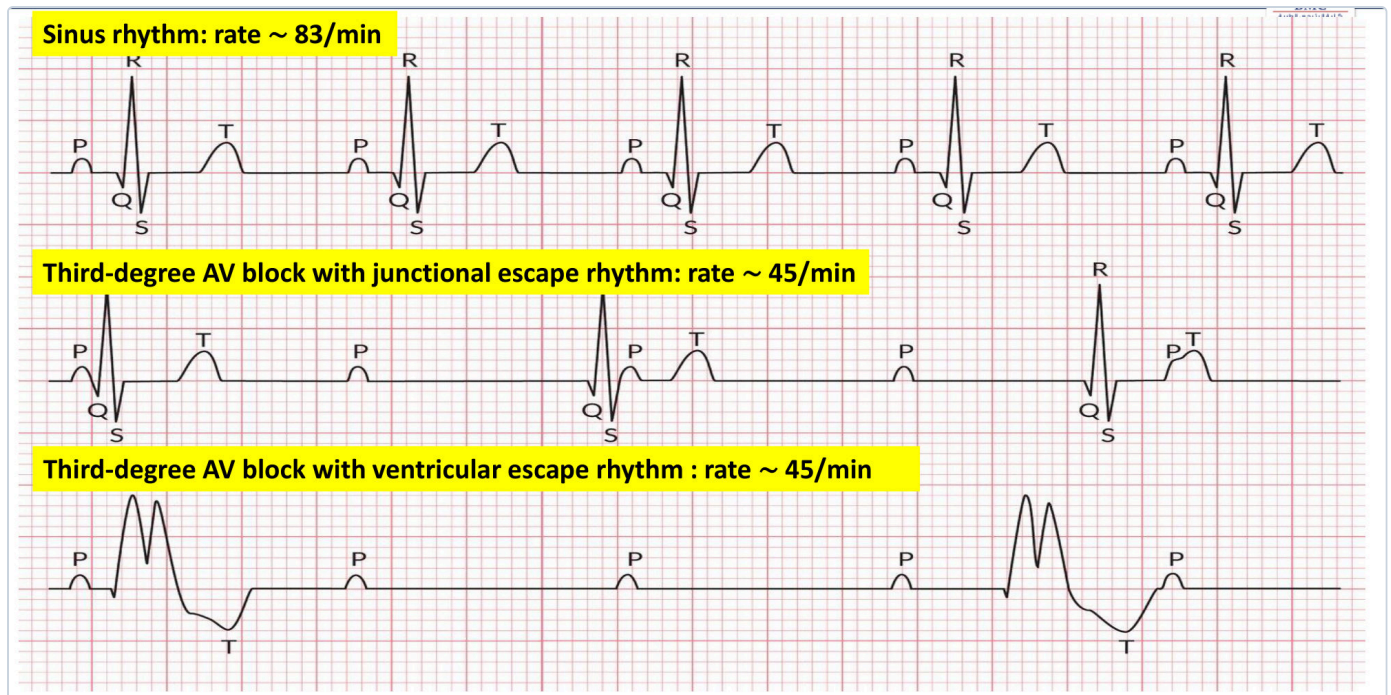
Block	ECG
1st degree	PR > 0.20 s , constant; every P conducted
2nd: Mobitz I (Wenckebach)	PR progressively lengthens until a P is dropped ; then the cycle resets
2nd: Mobitz II	PR constant; sudden non-conducted P
High-grade block	≥ 2 consecutive non-conducted P waves
3rd degree (complete)	P waves and QRS independent (AV dissociation) . Slide: sinus rate ~83 with a junctional escape ~45 (narrow) or a ventricular escape ~45 (wide)



Slide 19: first-degree AV block



Slide 20: Mobitz I, Mobitz II and high-grade block



Slide 21: complete heart block with junctional vs ventricular escape

AV block: follow the PR interval and the dropped QRS

Degree	PR / pattern (P = blue, QRS = black)	Level · danger · treatment
1st degree every P conducts	 PR constant > 0.20 s	AV node · benign No treatment; review drugs (β-blocker, CCB, digoxin)
2nd Mobitz I (Wenckebach)	 PR lengthens → P not conducted → resets	AV node · usually benign (inferior MI, vagal) Symptomatic → atropine Grouped beating
2nd Mobitz II (± high-grade)	 PR CONSTANT, sudden dropped QRS (wide QRS often)	His-Purkinje · can → complete block Permanent pacemaker Atropine often ineffective → pace
3rd degree (complete)	 P and QRS independent (AV dissociation); escape ~45/min	Junctional (narrow) or ventricular (wide) escape Cannon a waves · syncope (Stokes-Adams) Atropine → pacing → PERMANENT PACEMAKER

Unstable bradycardia: atropine 1 mg IV q3-5 min (max 3 mg) → transcutaneous/transvenous pacing or dopamine/adrenaline infusion

Added diagram: AV blocks compared

2.5 Management of Bradycardia (Slides 22-23) ★

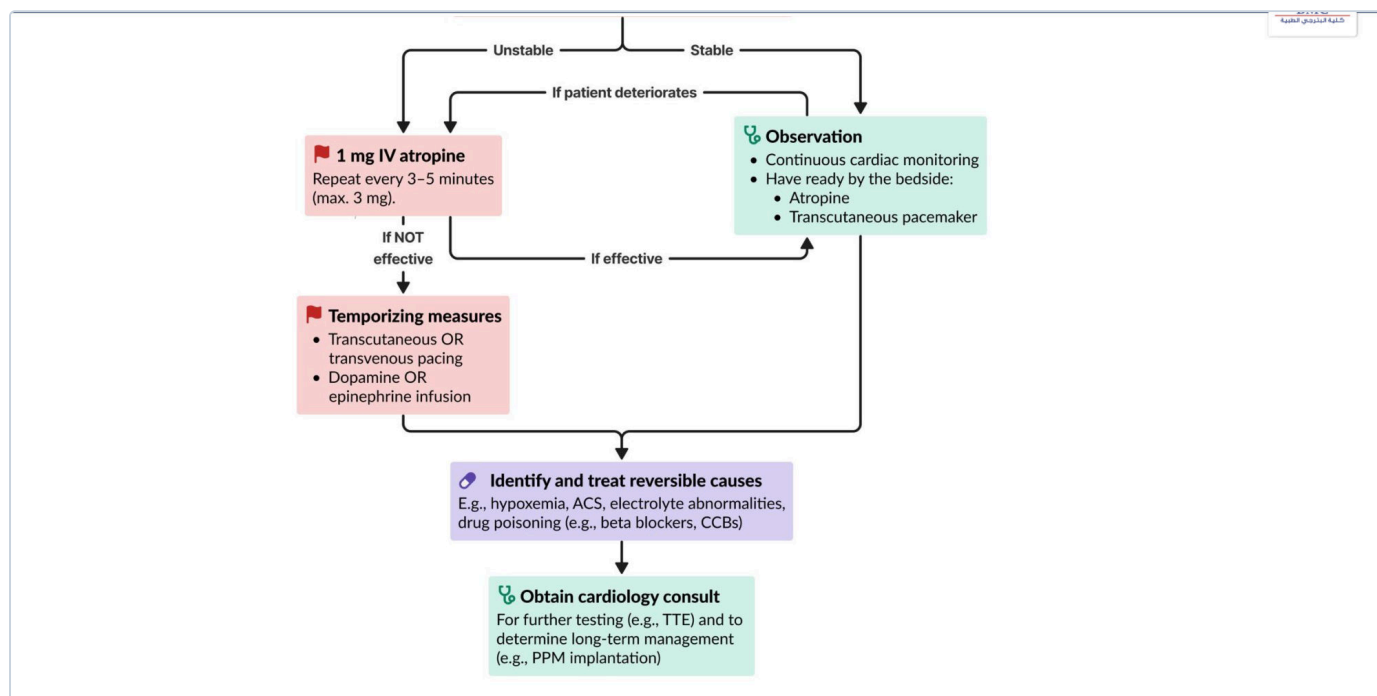
- Is the rhythm appropriate for the clinical condition?** (HR typically < 50/min if a bradyarrhythmia is the problem.)
- Identify and treat the cause.** - **Airway** and **breathing**: give **O₂** if short of breath. - **Cardiac monitor**, BP, oximetry. - **IV access**. - **12-lead ECG** (don't delay therapy).
- Persistent bradyarrhythmia causing any of these?** - **hypotension** - **acutely altered mental status** - **shock** - **ischaemic chest discomfort** - **acute heart failure**

No → monitor and observe.

Unstable	Stable
★ Atropine 1 mg IV , repeat every 3-5 min (max 3 mg)	Observation : continuous monitoring; atropine and transcutaneous pacemaker ready at the bedside
If not effective → temporising measures: transcutaneous OR transvenous pacing , OR dopamine / epinephrine infusion	If the patient deteriorates → atropine pathway

Then:

- **identify and treat reversible causes** (hypoxaemia, **ACS**, electrolytes, **β-blocker/CCB poisoning**)
- **cardiology consult** (TTE; long-term **PPM**)



Slide 23: bradycardia algorithm

[added] Atropine is often ineffective in Mobitz II/3rd degree with wide escape and in heart transplant patients (denervated heart), so go straight to pacing. Antidotes: glucagon for β-blocker overdose, calcium for CCB overdose, digoxin Fab for digoxin toxicity.

🔁 **Recap: Bradyarrhythmias.** Sinus arrhythmia is normal. Sinus brady, pauses and tachy-brady (SSS). AV blocks: 1st (PR > 0.2), Mobitz I (PR lengthens then drops, benign), Mobitz II (constant PR then drops, dangerous), 3rd (dissociation, escape ~45). If unstable (↓BP, confusion, shock, chest pain, HF): atropine 1 mg q3-5 min (max 3 mg) → pacing or dopamine/adrenaline → find the cause → PPM.

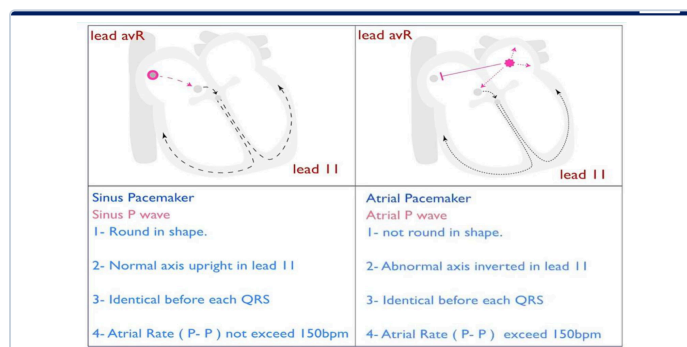
🔗 **Link:** At the other end of the rate spectrum, the first job is to recognise which tachycardia it is.

👉 **Up next: Tachyarrhythmia ECG gallery.**

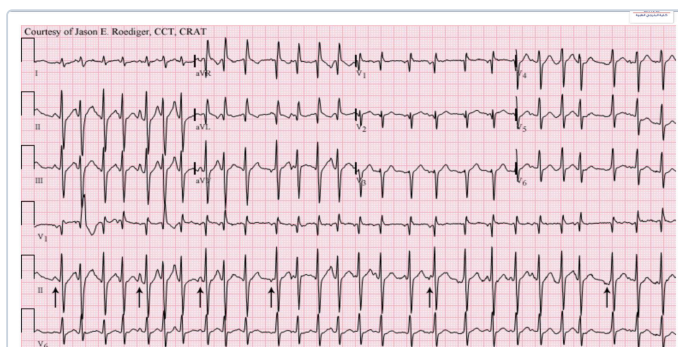
2.6 Tachyarrhythmia ECG Gallery (Slides 26-39)

Rhythm	ECG key
Sinus tachycardia	Normal P before each QRS, rate > 100 (usually ≤ 150 at rest [added])

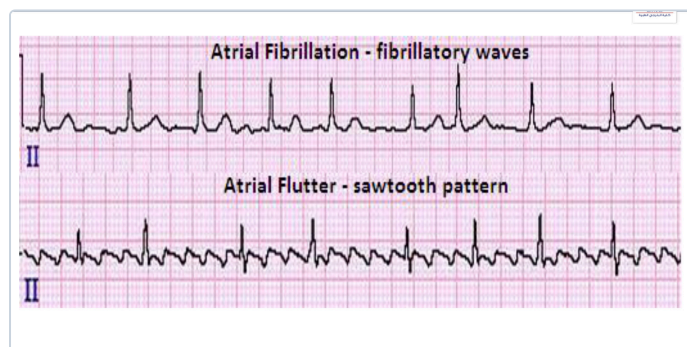
Rhythm	ECG key
Sinus vs focal atrial tachycardia (slide 27)	Sinus P: round, upright in II , normal axis, atrial rate ≤ 150 . Atrial focus P: not round, abnormal axis (inverted in II) , rate > 150
SV premature beats	Early, abnormal P-QRS with a normal QRS
★ MAT	≥ 3 different P-wave morphologies , irregular, rate > 100 ([added] COPD, hypoxia, theophylline)
★ Atrial flutter	★ Sawtooth flutter waves (lead II), atrial $\sim 300/\text{min}$, often 2:1 → ventricular rate ~ 150 [added]
★ AF	No P waves; fibrillatory baseline; irregularly irregular QRS
AF with rapid ventricular response	AF with a ventricular rate $> 100-110$
PVCs	Unifocal (identical) vs multifocal (> 1 shape); wide, bizarre, early, compensatory pause [added]
Ventricular bigeminy	Every other beat is a PVC
★ Monomorphic VT	Regular wide-complex tachycardia, identical QRS (slide example HR 182)
★ Torsades de pointes	Polymorphic VT twisting around the baseline
★ VF	Chaotic, no QRS. Fine VF < 3 mm vs coarse VF > 3 mm



Slide 27: sinus vs focal atrial tachycardia P waves



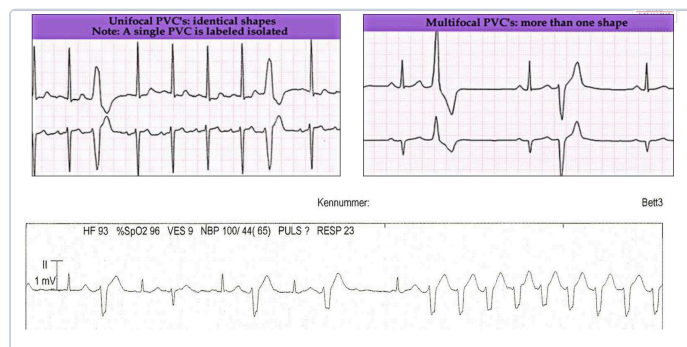
Slide 29: multifocal atrial tachycardia



Slide 30: AF (fibrillatory waves) vs flutter (sawtooth)



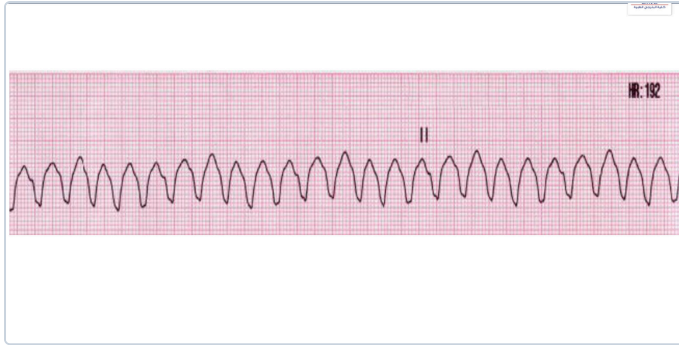
Slide 31: atrial fibrillation



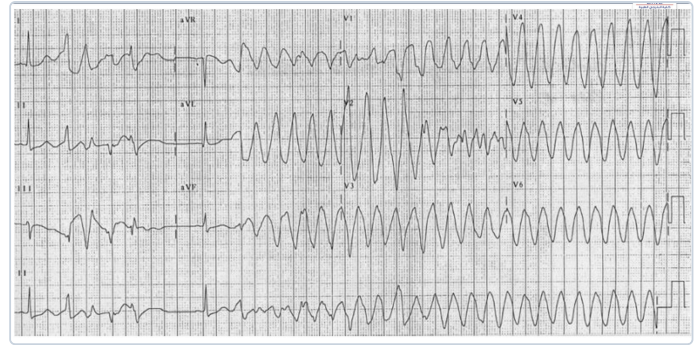
Slide 33: unifocal vs multifocal PVCs → VT



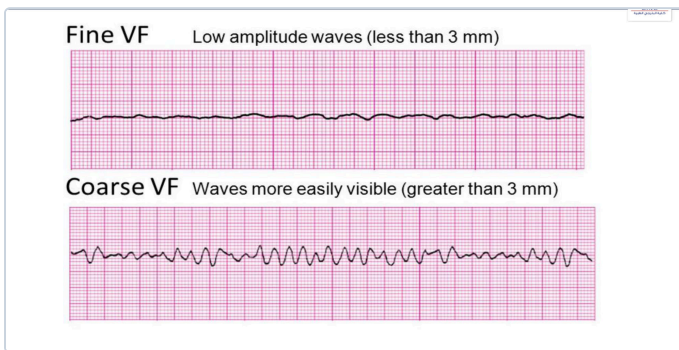
Slide 34: ventricular bigeminy



Slide 36: monomorphic VT



Slide 37: torsades de pointes



Slide 38: fine vs coarse VF



Slide 73: monomorphic vs polymorphic VT vs torsades

PVC treatment (slide 34):

- **Most patients (asymptomatic) need none.**
- **Treat the underlying disease.**
- Treat only **frequent and significantly symptomatic** PVCs, with **β -blockers**.

2.7 Tachycardia Management Framework (Slides 40-43) ★

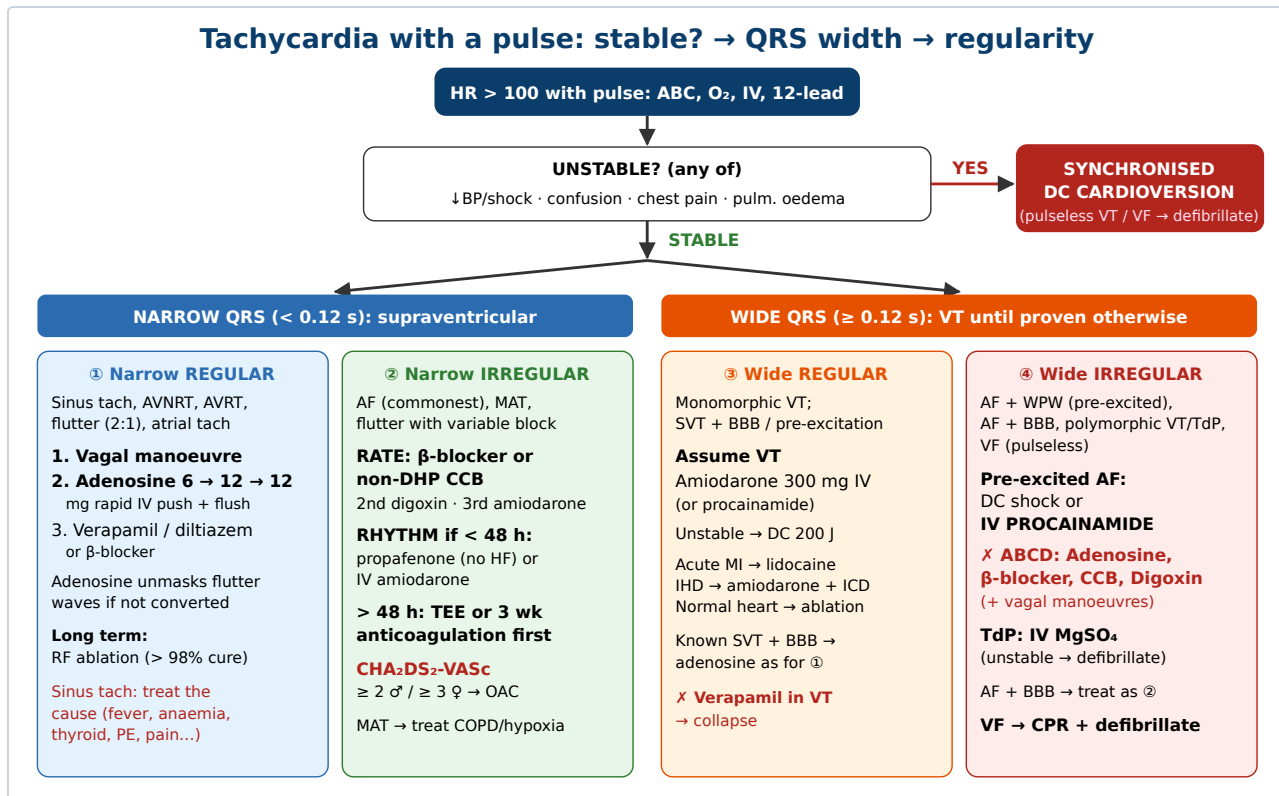
Step 1: stable or unstable?

- **Signs of haemodynamic instability:** - chest pain - altered mental status - acute pulmonary oedema - hypotension - cardiogenic shock
- ★ **Unstable → DC cardioversion.**

Step 2 (stable): regular or irregular × narrow or wide → four groups

	Narrow QRS	Wide QRS
Regular	Sinus tach, SVT (AVNRT/AVRT), atrial flutter, atrial tachycardia	VT (or SVT + accessory pathway / BBB)
Irregular	AF, MAT	AF + (accessory pathway or BBB), VF

Tachycardia with a pulse: stable? → QRS width → regularity



Added diagram: tachycardia algorithm by stability, QRS width and regularity

Recap: Tachyarrhythmia types. Sinus vs atrial tachycardia by P shape and axis. MAT = ≥ 3 P morphologies. Flutter = sawtooth. AF = no P, irregularly irregular. PVCs are benign unless frequent and symptomatic (β-blocker). VT = regular wide. TdP = twisting polymorphic. VF = chaos. Management: unstable → DC shock; stable → sort into narrow/wide × regular/irregular.

Link: The four groups, one at a time, starting with narrow regular (SVTs).

Up next: Narrow regular tachycardia.

2.8 Narrow Regular: Sinus Tachycardia, AVNRT, AVRT (Slides 44-58)

Causes:

- **sinus tachycardia**
- **AVNRT**
- **AVRT**
- **atrial flutter**
- **atrial tachycardia**

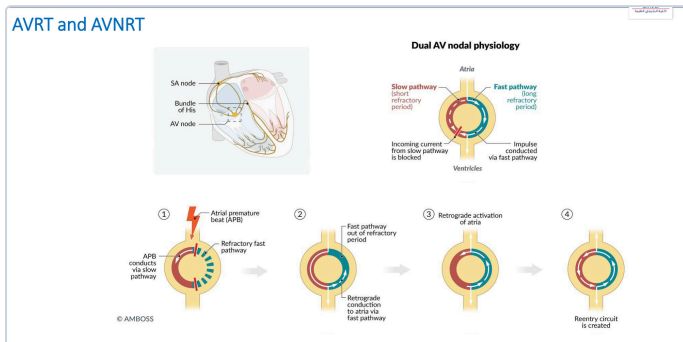
Sinus tachycardia treatment (slide 46):

- **Asymptomatic** without structural heart disease → **no treatment**.
- **Symptomatic:** - reduce triggers (**caffeine, alcohol, stress, smoking**) - treat **correctable causes** (**anaemia, thyrotoxicosis, fever**) - **β-blockers** for persistent symptoms

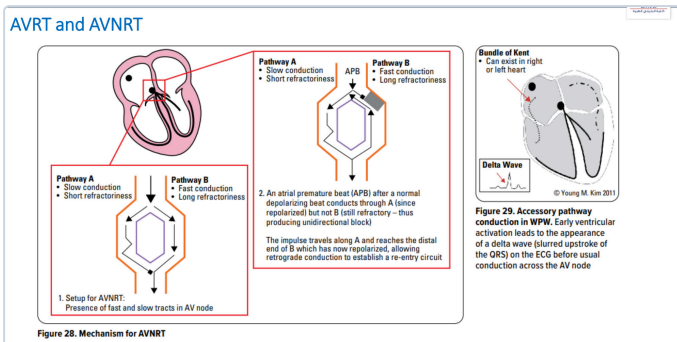
AVNRT (slides 47-49) ★

- **Mechanism:** - **Re-entry using dual pathways** within or near the AV node: **fast** (β-fibres; long refractory period) and **slow** (α-fibres; short refractory period). - It is usually triggered by a **premature beat**: the beat goes down the slow pathway (the fast one is still refractory), then back up the fast pathway, which creates a re-entry circuit.

- **Usually a structurally normal heart**; idiopathic (familial cases reported).
- **Sudden onset and offset**; ★ "neck pounding" ([added] frog sign: atria contract against closed AV valves).
- **Regular, 150-250 bpm.**
- ★ **60-70% of all paroxysmal SVTs** (the commonest).
- **Retrograde P waves are usually lost in the QRS** (or just after it; [added] pseudo-r' in V1).



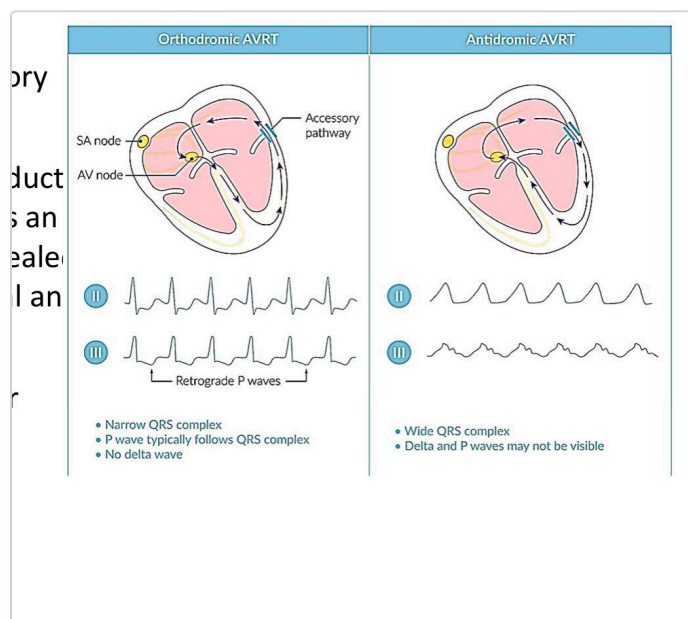
Slide 47: dual AV nodal pathway physiology



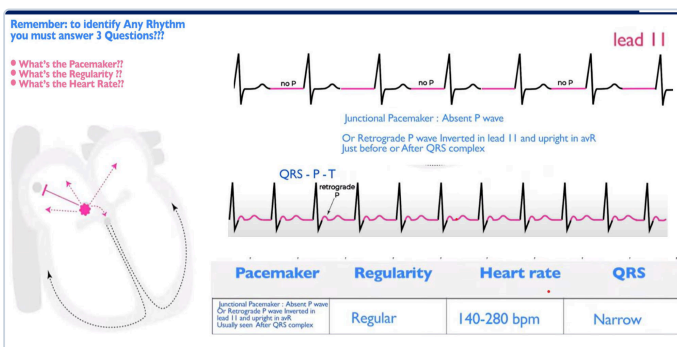
Slide 48: AVNRT mechanism; delta wave in WPW

AVRT (slide 50) ★

- **Re-entrant loop**: down the **normal conduction system (antegrade)** and **up an accessory pathway (retrograde)** = ★ **orthodromic (narrow QRS)**. - The reverse direction (down the accessory pathway) = **antidromic (wide QRS)**.
- Usually in patients with an **antegradely conducting bypass tract (WPW)**.
- With a **concealed** (retrograde-only) pathway, the resting **ECG is normal, with no delta waves**.
- **Triggered by a premature atrial or ventricular complex.**



Slide 50: orthodromic vs antidromic AVRT



Slide 53: PSVT (AVNRT) rhythm analysis

Slide 53 analysis: pacemaker **junctional** (absent P, or retrograde P inverted in II and upright in aVR, just before or after the QRS) · **regular** · **140-280 bpm** · **narrow** QRS.

Treatment (slides 51-52, 54-58) ★

	AVNRT	AVRT
Unstable	Electrical cardioversion	Same

	AVNRT	AVRT
Acute, stable	1. Vagal manoeuvres : Valsalva or carotid sinus pressure (<i>with precautions</i>). 2. ★ Adenosine 6 mg rapid IV (incremental doses) = first choice if vagal fails . 3. Verapamil (Isoptin) 5 mg IV over 3-5 min , repeated. 4. If no response: metoprolol, digoxin, diltiazem	As for AVNRT , but ★ avoid long-acting AV-nodal blockers (digoxin, verapamil)
Long-term	★ 1st line: radiofrequency ablation (> 98% cure, < 1% complications); or β -blocker, diltiazem, digoxin. 2nd line: flecainide, propafenone	★ Ablation of the bypass tract for recurrence; drugs: flecainide, procainamide



Slide 54: SVT 12-lead

[added]:

- Adenosine doses: **6 → 12 → 12 mg** rapid push with a flush, through a proximal (antecubital) vein.
- Warn the patient about **transient chest discomfort / sense of doom**.
- **Contraindicated in asthma** (bronchospasm).
- **Dipyridamole** potentiates adenosine; **theophylline** blocks it.
- Carotid massage is **contraindicated with a carotid bruit or a recent TIA/stroke**. These are the "precautions" the slide refers to.

After reversion (slide 57):

- **ECG normal** → maintenance with a **β -blocker or verapamil**.
- **WPW pattern** → ★ **refer for EP study + RF ablation**.

RF ablation: tissue destruction by **radiofrequency current (~80 °C)**. Success **97% vs 60% with amiodarone** (slide 58).

🔗 **Recap: Narrow regular**. Sinus tach → treat the cause. AVNRT is the commonest PSVT (60–70%): dual AV-nodal pathways, sudden onset, neck pounding, 150–250 bpm, P hidden in the QRS. AVRT uses an accessory pathway (orthodromic narrow; WPW). Treatment: unstable → DC; stable → vagal → adenosine 6 mg rapid → verapamil 5 mg / β -blocker / diltiazem. In AVRT avoid long-acting nodal blockers. Cure = RF ablation (> 98%); WPW → EP study + ablation.

🔗 **Link**: If the narrow rhythm is irregular, it is almost always AF.

👉 **Up next: Atrial fibrillation**.

2.9 Narrow Irregular: Atrial Fibrillation (Slides 59–68) ★

Causes: most commonly **AF** (also MAT, flutter with variable block [added]).

AF classification (slide 59):

Basis	Type	Definition
Ventricular rate	AF with rapid ventricular response	Ventricular rate > 100-110/min (tachycardic AF)
	AF with slow ventricular response	< 60/min (bradycardic AF)
Onset and duration	New-onset	First presentation, regardless of duration

Basis	Type	Definition
	Paroxysmal	Resolves within 7 days (spontaneously or with treatment); may recur
	Persistent	Continuous > 7 days
	Long-standing persistent	Continuous > 1 year
	Permanent	Persistent AF where no further attempts at sinus rhythm are planned (patient and physician agree)

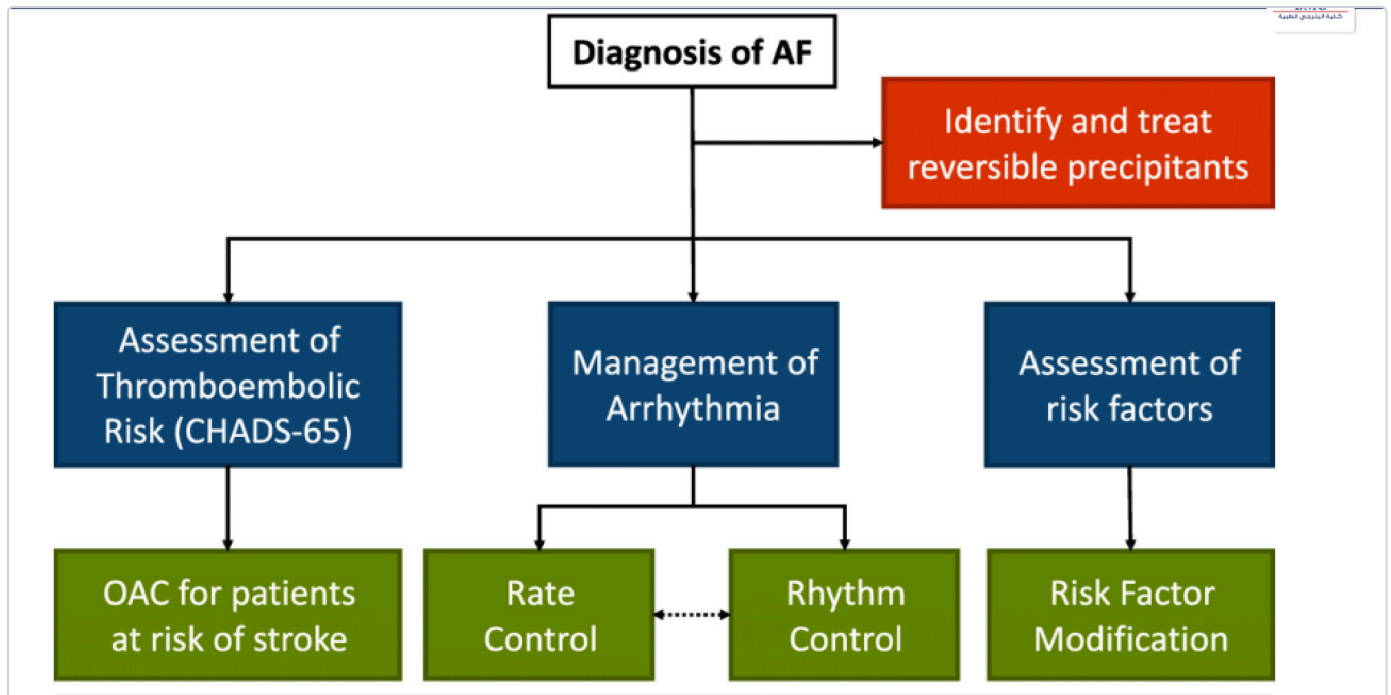
- A-fib classification

Ventricular rate	Afib with rapid ventricular response	Afib with a ventricular rate > 100–110/minute (tachycardic Afib) ^{[10][11]}
	Afib with slow ventricular response ^[12]	Afib with a ventricular rate < 60/minute (bradycardic Afib or slow Afib)
Onset and duration	New-onset Afib	New presentation or diagnosis of Afib, regardless of duration ^[13]
	Paroxysmal Afib	Afib that resolves within 7 days of onset either following treatment or spontaneously; the frequency of recurring episodes may vary.
	Persistent Afib	Continuous Afib for > 7 days
	Long-standing persistent Afib	Continuous Afib for > 1 year
	Permanent Afib	Persistent Afib in which therapeutic attempts are no longer made to convert to or maintain <u>sinus rhythm</u> unless the patient and the treating physician agree to do so ^[5]

Slide 59: AF classification

Management overview (slide 68):

- Diagnosis of AF.**
- Identify and treat reversible precipitants.**
- Three parallel assessments: - **thromboembolic risk (CHA₂DS₂-VASc)** → **OAC** for those at risk - **arrhythmia management** → **rate** and/or **rhythm control** - **risk factors** → **risk-factor modification**



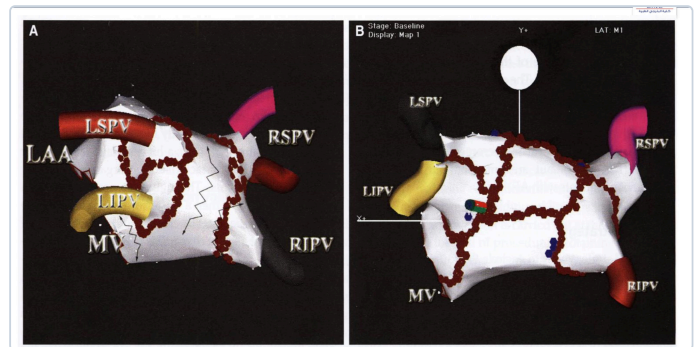
Slide 68: AF management overview

Rhythm control (slides 60-63):

Situation	Management
★ Haemodynamically unstable	DC shock 150 J (synchronised, biphasic)
Stable, recent onset (< 48 h)	Medical reversion: oral propafenone 600 mg (only if no CHF / impaired LV function), OR IV amiodarone: loading 300 mg IV over 40 min, then 1800 mg/24 h infusion
★ AF > 48 h	Exclude LA thrombus by TEE, and/or anticoagulate before reversion. Anticoagulate after reversion with warfarin (INR 2-3) ([added] ≥ 4 weeks; the atria are stunned)
Maintenance after cardioversion	Amiodarone 200 mg/day or sotalol 80 mg bid
Frequent recurrence	★ Pulmonary vein isolation (catheter ablation around the PV ostia)



Slide 62: TEE showing LA appendage thrombus



Slide 63: pulmonary vein isolation map

Rate control (slide 64):

- **First line:** ★ **β-blockers** or **non-DHP CCBs** (verapamil, diltiazem; *[added]* avoid in HFrEF)
- **Second line:** **digoxin** (*[added]* useful in HF or sedentary patients)
- **Third line:** **amiodarone**
- **Non-drug:** **AV-node ablation + permanent ventricular pacemaker** ("ablate and pace")
- ★ **"Do not forget to anticoagulate": INR 2-3**

Stroke prevention: CHA₂DS₂-VASc (slide 65) ★

Letter	Condition	Points
C	Congestive HF / LV systolic dysfunction	1
H	Hypertension (> 140/90 or treated)	1
A₂	Age ≥ 75	2
D	Diabetes	1
S₂	Stroke / TIA / thromboembolism	2
V	Vascular disease (PAD, MI, aortic plaque)	1
A	Age 65-74	1
Sc	Sex category: female	1

CHA₂DS₂ (the older score) has only C, H, A (≥ 75 = 1), D, S₂. CHA₂DS₂-VASc refines patients scoring 0-1 on CHA₂DS₂ and identifies those at truly low risk.

CHA₂DS₂-VASc score

مراجعة
مراجعة

Condition	CHADS ₂ score	Points	CHA ₂ DS ₂ -VASc score	Points
Congestive heart failure (or Left ventricular systolic dysfunction)	C	1	C	1
Hypertension: blood pressure consistently above 140/90 mmHg (or treated hypertension on medication)	H	1	H	1
Age ≥75 years	A	1	A ₂	2
Diabetes Mellitus	D	1	D	1
Stroke or TIA or thromboembolism in history	S ₂	2	S ₂	2
Vascular disease (e.g. peripheral artery disease, myocardial infarction, aortic plaque)			V	1
Age 65–74 years			A	1
Sex category (i.e. female gender)			Sc	1

This table shows the components of the CHADS₂ (Gage et al., JAMA 2001 [36]) and CHA₂DS₂-VASc scores (Lip et al., Chest 2010 [41]) tools to assess stroke risk in patients with AF. These risk assessment tools help to determine who should and who should not receive anticoagulation. CHA₂DS₂-VASc improves risk stratification in patients with CHADS₂=0 or 1, and allows for identification of patients at truly low risk.

Slide 65: CHADS₂ vs CHA₂DS₂-VASc

Who to anticoagulate (slide 66) ★

Risk	Criteria	Action
High	★ Valvular disease ([added] moderate-severe MS or mechanical valve), HCM , and/or CHA₂DS₂-VASc ≥ 2 (men) / ≥ 3 (women)	Long-term anticoagulation
Medium	Non-valvular AF with a score of 1 (men) / 2 (women)	Consider anticoagulation
Low	Non-valvular AF with a score of 0 (men) / 1 (women)	Not routinely recommended

★ **Always weigh the bleeding risk before starting anticoagulation.**

Bleeding risk: HAS-BLED (slide 67)

Letter	Item	Points
H	Hypertension, uncontrolled (SBP > 160)	1
A	Abnormal renal (dialysis, transplant, Cr > 200 μmol/L) and/or liver function (cirrhosis, bilirubin > 2× ULN, AST/ALT/ALP > 3× ULN)	1 each (1–2)
S	Stroke history	1
B	Bleeding history or predisposition	1
L	Labile INR (TTR < 60%)	1
E	Elderly (> 65)	1
D	Drugs (antiplatelets, NSAIDs) and/or alcohol (≥ 8 drinks/week)	1 each (1–2)

Risk bands on the slide: **0 = low** • **1-2 = moderate** • **≥ 3 = high**.

Clinical characteristic			Points
Hypertension	Uncontrolled, >160 mmHg systolic)	H	1
Abnormal renal and liver function (1 point each)	Dialysis, transplant, Cr >200 µmol/L, Cirrhosis, bilirubin >2x normal, AST/ALT/AP >3x normal)	A	1 or 2
Stroke history		S	1
Bleeding or predisposition to bleeding		B	1
Labile INR	Unstable/high INRs, time in therapeutic range < 60%)	L	1
Elderly	Age > 65	E	1
Drugs or alcohol (1 point each)	Antiplatelet agents, NSAIDs, ≥ 8 alcohol drinks/week	D	1 or 2

This table shows the components of the HAS-BLED score (Pisters et al, Chest 2010 [45], Camm et al., Eur H J 2010 [2]), used to assess bleeding risk.

AF patients are subdivided into 3 risk stratifications,

- a score of 0 indicates low risk
- 1–2 indicates moderate risk
- ≥3 indicates high risk

Slide 67: HAS-BLED

[added] A high HAS-BLED means correct the modifiable bleeding factors and monitor closely. It is not by itself a reason to withhold OAC. Non-valvular AF → DOACs preferred over warfarin. Mechanical valves / moderate-severe MS → warfarin only.**

Recap: AF.

- **Classification:** rapid vs slow response; new-onset, paroxysmal (≤ 7 d), persistent (> 7 d), long-standing (> 1 y), permanent.
- **Unstable:** DC 150 J.
- **< 48 h:** propafenone (no HF) or IV amiodarone 300 mg then 1800 mg/24 h.
- **> 48 h:** TEE / anticoagulate first, warfarin INR 2–3 after. Maintain with amiodarone 200 or sotalol 80 bid; recurrent → PV isolation.
- **Rate control:** β-blocker / non-DHP CCB → digoxin → amiodarone → ablate and pace.
- **Stroke prevention:** CHA₂DS₂-VASc ≥ 2 ♂ / ≥ 3 ♀ (or valvular, HCM) → OAC; weigh HAS-BLED.

 **Link:** Wide regular means VT until proven otherwise.

 **Up next: Wide-complex tachycardias.**

2.10 Wide Regular: VT (Slides 69–73) ★

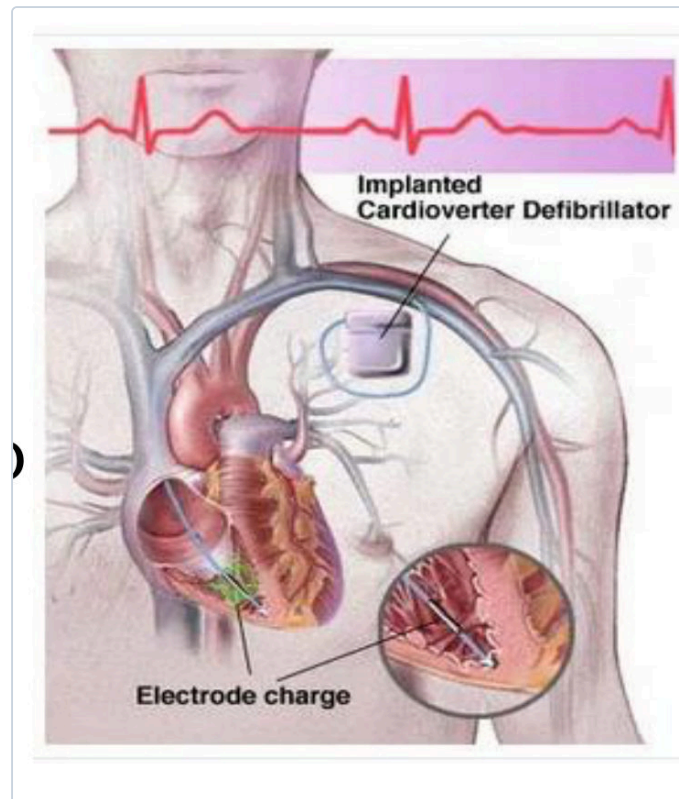
Causes: VT or **SVT with aberration** ([added] BBB, pre-excitation).

Definition:

- **VT = ≥ 3 consecutive ventricular complexes (wide QRS) at ≥ 100/min.**
- **Non-sustained VT (NSVT): < 30 s. Sustained VT: ≥ 30 s.**
- **Monomorphic:** identical QRS, from **a single focus**. **Polymorphic:** varying QRS, from **multiple foci**.

VT management (slide 69):

Setting	Treatment
★ Unstable	DC shock 200 J
Complicated acute MI	Lidocaine (Xylocaine)
Complicated IHD	★ Amiodarone + ICD (AICD), or RF ablation (low success rate)
Normal heart (young)	★ RF ablation ([added] idiopathic RVOT/fascicular VT)



Slide 69: implanted cardioverter defibrillator

[added] Stable VT: amiodarone 300 mg IV over 20–60 min, then 900 mg/24 h (slide 84), or procainamide. Clues that a wide-complex tachycardia is VT, not SVT with aberrancy:

- AV dissociation, capture or fusion beats
- $QRS > 0.16$ s
- extreme axis
- concordance in V1–V6
- history of IHD

Never give verapamil to a wide-complex tachycardia.

Torsades de pointes (slide 72) ★

- **Etiology:** congenital long QT syndrome or acquired LQTS: drugs, electrolyte abnormalities ([added] ↓K, ↓Mg, ↓Ca), heart disease.
- **ECG:** polymorphic VT with QRS twisting around the isoelectric line.
- **Acute management:** - ★ Unstable → defibrillation + CPR - ★ Stable → IV magnesium sulfate ([added] 2 g over 10 min)
- **Complication:** can progress to VF.

[added] QT-prolonging drugs:

- class IA/III antiarrhythmics
- macrolides, fluoroquinolones
- haloperidol, antipsychotics

- TCAs, methadone, ondansetron

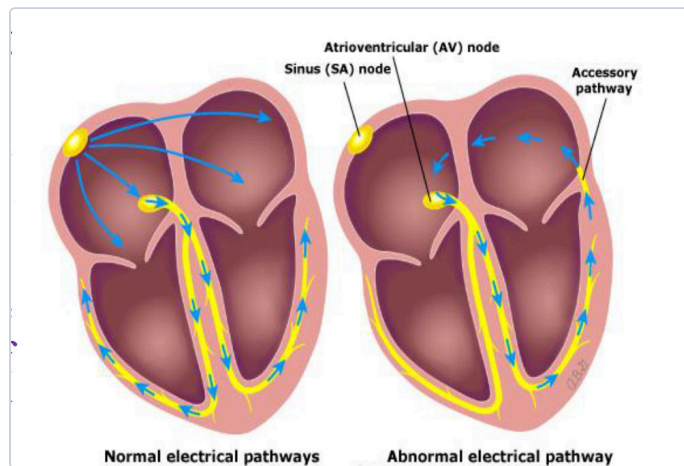
Also: stop the offending drug, correct K^+ , and use overdrive pacing / isoprenaline if recurrent.

2.11 Wide Irregular: VF and Pre-excited AF (Slides 74-77) ★

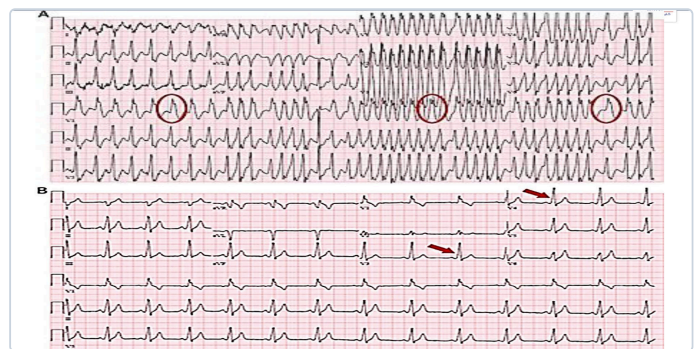
Causes: ★ **VF** or **AF complicating WPW**. Both are usually **haemodynamically unstable** → **DC cardioversion** (VF needs [added] **unsynchronised** defibrillation + CPR).

★ **All drugs that block the AV node are CONTRAINDICATED in AF with an accessory pathway (WPW).**

- [added] Reason: blocking the node pushes all impulses down the accessory pathway, which can conduct at up to 300/min → **VF**.



Slide 74: normal vs WPW conduction pathways



Slide 76: AF + WPW (irregular, varying wide QRS)

WPW treatment (slide 77):

- **Unstable** → **electrical cardioversion**.
- **Stable** → assess the underlying rhythm. - **Wide-complex tachycardia: irregular** (underlying AF or MAT) or **regular** (underlying flutter or focal AT). - ★ **Avoid AV-nodal blocking agents AND vagal manoeuvres**. - ★ **Rhythm control (cardioversion or IV procainamide) is the safest option**.
- [added] Then **EP study + ablation of the pathway**.

Mnemonic: in WPW + AF avoid "ABCD" [added] = **A**denosine · **B**eta-blockers · **C**alcium-channel blockers (verapamil, diltiazem) · **D**igoxin (± amiodarone IV in some guidance).

🔄 Recap: Wide tachycardias.

- **Wide regular** = VT until proven otherwise: unstable → DC 200 J; acute MI → lidocaine; IHD → amiodarone + ICD; normal heart → ablation.
- **TdP**: long QT; stable → IV Mg; unstable → defibrillate.
- **Wide irregular** = VF (CPR + defibrillate) or AF + WPW (DC shock or procainamide; never ABCD or vagal manoeuvres).

🔗 **Link:** The non-drug tools used so far, in one place.

👉 **Up next: Non-pharmacological therapy and drug classes.**

2.12 Non-Pharmacological Management (Slides 78-83)

Summary of tools (slide 78):

- **vagal manoeuvres**
- **drugs** (anti-tachycardia and anti-bradycardia agents)
- **cardioversion** (electrical and chemical)
- **RF catheter ablation**
- **pacemaker therapy**

Vagal manoeuvres (slides 79-80) increase parasympathetic tone. They are used to **abort stable SVT** or **differentiate SVT from VT**.

Manoeuvre	Technique
Carotid sinus massage	Supine , neck extended and rotated to the opposite side . Carotid sinus = pulse just below the angle of the mandible , anterior to SCM, at the upper level of the thyroid cartilage . Index + middle fingertips press posteromedially for 5-10 s . Repeat after 1 min if unsuccessful, then try the opposite side ([added] never both at once)
Valsalva	Supine , deep breath, blow against a closed glottis for 10-15 s ([added] modified Valsalva: then lie flat and raise the legs, which doubles success)

Cardioversion (slide 82):

1. **Electrical cardioversion**: a high-energy shock resets the rhythm. - ★ **Synchronised** (timed to the R wave; [added] AF, flutter, SVT, VT with pulse). - **Unsynchronised = defibrillation**: for life-threatening rhythms, especially **VF** and **pulseless VT**. - ★ **Asystole cannot be restarted by a defibrillator** → **CPR** ([added] + adrenaline).
2. **Chemical cardioversion**: drugs restore sinus rhythm.

Interventional therapy (slide 83): radiofrequency catheter ablation.

2.13 Antiarrhythmic Drug Classes (Slide 81)

Vaughan Williams classification:

Class	Mechanism	Drugs
IA	Block fast Na⁺ channels (moderate)	Quinidine, procainamide, disopyramide , moricizine
IB	Na⁺ (weak; ischaemic tissue)	Lidocaine , tocainide, mexiletine
IC	Na⁺ (strong)	Flecainide, propafenone
II	β-blockers (↓ depolarisation rate)	Esmolol, propranolol , acebutolol
III	Block K⁺ channels (prolong repolarisation)	★ Amiodarone, sotalol , ibutilide, dofetilide, bretylium
IV	Block slow Ca²⁺ channels	Verapamil, diltiazem
Misc.		Adenosine, digoxin, magnesium, potassium

By site of action:

Supraventricular only	Supraventricular + ventricular	Ventricular only
Adenosine, verapamil, diltiazem, dronedarone, digoxin	Amiodarone, β-blockers (propranolol, sotalol, esmolol), procainamide, disopyramide, quinidine, flecainide, propafenone	Lidocaine, mexiletine

Pro-arrhythmic effects:

- **Class Ia, Ic**: prolong QT; can cause **VT/VF in CAD and HF** ([added] CAST trial: flecainide after MI ↑ mortality)
- **Class III**: like Ia/Ic (QT → TdP)

- **Class II, IV: bradycardia**

Supraventricular Arrhythmias Only	Supraventricular and ventricular arrhythmias		Ventricular arrhythmias only	Table 21-1 Classification of Cardiac Antiarrhythmic Drugs Class I (inhibit fast sodium ion channels) Class IA Quinidine Procainamide Disopyramide Moricizine Class IB Lidocaine Tocainide Mexiletine Class IC Flecainide Propafenone Class II (decrease rate of depolarization) Esmolol Propranolol Acebutolol Class III (inhibit potassium ion channels) Amiodarone Sotalol Ibutilide Dofetilide Bretylium Class IV (inhibit slow calcium channels) Verapamil Diltiazem Miscellaneous agents Adenosine Digoxin Magnesium Potassium
Adenosine	Amiodarone	Procainamide	Lidocaine	
Verapamil	β blockers	Disopyramide	Mexilitine	
Diltiazem	Propanolol	Quinidine		
Dronadarone	Sotalol	Flecainide		
Digoxin	Esmolol	Propafenone		

Proarrhythmia effect of antiarrhythmia agents

- ▮ Ia, Ic class: Prolong QT interval, will cause VT or VF in coronary artery disease and heart failure patients
- ▮ III class: Like Ia, Ic class agents
- ▮ II, IV class: Bradycardia

Slide 81: antiarrhythmic classes and pro-arrhythmia

Mnemonic: "Some Block Potassium Channels" [added] = class **I** Sodium · **II** Beta · **III** Potassium · **IV** Calcium. For subclasses: **IA "Double Quarter Pounder"** (Disopyramide, Quinidine, Procainamide) · **IB "Lettuce, Tomato, Mayo"** (Lidocaine, Tocainide, Mexiletine) · **IC "Fries Please"** (Flecainide, Propafenone).

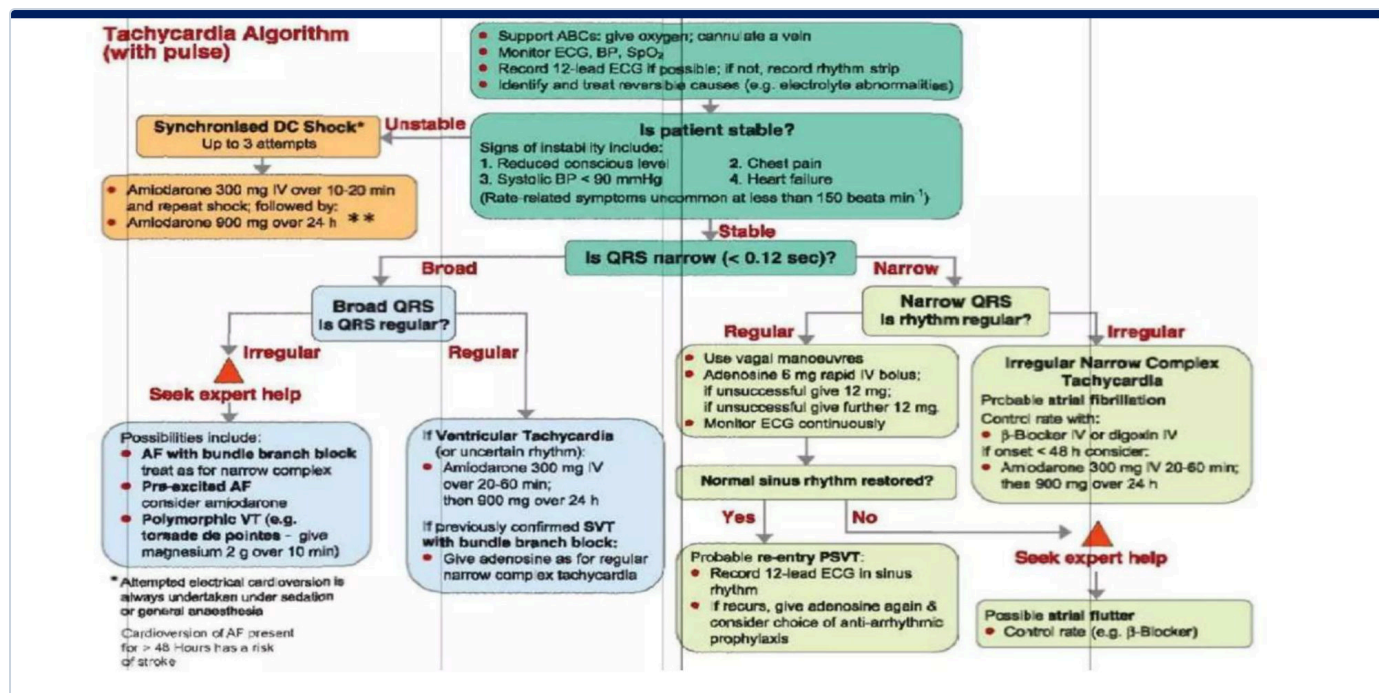
2.14 Summary Algorithm (Slide 84) ★

Tachycardia with a pulse:

- **Support ABCs:** O₂, IV, monitor ECG/BP/SpO₂, 12-lead, treat reversible causes.
- **Unstable** (↓ consciousness, chest pain, **SBP < 90**, heart failure; rate-related symptoms are uncommon < 150 bpm): - **synchronised DC shock, up to 3 attempts** (under sedation/anaesthesia) - then **amiodarone 300 mg IV over 10-20 min** and repeat the shock - then **amiodarone 900 mg over 24 h**
- **Stable → QRS narrow (< 0.12 s)?**

	Regular	Irregular
Narrow	Vagal manoeuvres → adenosine 6 mg rapid IV , then 12 mg , then 12 mg → if sinus restored = re-entry PSVT (record ECG; if it recurs give adenosine again and consider prophylaxis); if not restored = possible flutter → rate control (β-blocker)	Probable AF → rate control: IV β-blocker or digoxin ; if onset < 48 h consider amiodarone 300 mg over 20-60 min , then 900 mg/24 h
Broad	VT (or uncertain) → amiodarone 300 mg IV over 20-60 min , then 900 mg/24 h . Known SVT + BBB → adenosine as for narrow	Seek expert help . Possibilities: AF + BBB (treat as narrow) · pre-excited AF (consider amiodarone) · polymorphic VT/TdP → magnesium 2 g over 10 min*

Cardioversion of AF present > 48 h carries a stroke risk.



Slide 84: tachycardia algorithm (with pulse)

3. High-Yield Comparison Tables

Table A: The 4 Tachycardia Groups

Group	Differential	First-line (stable)	Contraindicated
Narrow regular	Sinus tach, AVNRT , AVRT (ortho), flutter 2:1 , atrial tach	Vagal → adenosine → verapamil/β-blocker	Adenosine in asthma [added]
Narrow irregular	AF , MAT, flutter with variable block	Rate: β-blocker / non-DHP CCB; anticoagulate	Propafenone if HF
Wide regular	VT , SVT + BBB, antidromic AVRT	Amiodarone / procainamide ; DC if unstable	★ Verapamil
Wide irregular	VF , AF + WPW , AF + BBB, TdP	DC; procainamide (pre-excited AF); Mg (TdP)	★ AV-nodal blockers + vagal (WPW)

Table B: AVNRT vs AVRT

	AVNRT	AVRT
Circuit	Dual pathways in/near the AV node	AV node + accessory pathway
Frequency	60-70% of PSVT (commonest)	Less common; WPW
Resting ECG	Normal	Short PR + delta wave (unless concealed)
Tachycardia QRS	Narrow	Orthodromic narrow / antidromic wide
P waves	Hidden in QRS	Retrograde P after QRS [added]
Acute treatment	Vagal → adenosine → verapamil/β-blocker	Same, but avoid long-acting nodal blockers (digoxin, verapamil)
Cure	RF ablation (slow pathway)	Ablation of the bypass tract

Table C: AV Blocks

	PR	Dropped beats	Site	Risk	Treatment
1st	> 0.20, constant	None	AV node	Benign	None
Mobitz I	Progressive ↑	Periodic	AV node	Low	Atropine if symptomatic
Mobitz II	Constant	Sudden	Infranodal	High → CHB	Pacemaker
3rd	No relation	AV dissociation	Any	High	Pacemaker

Table D: AF: Rate vs Rhythm

	Rate control	Rhythm control
Drugs	β-blocker, non-DHP CCB → digoxin → amiodarone	Propafenone (no HF), amiodarone , sotalol
Procedure	AV-node ablation + PPM	DC cardioversion (150 J), PV isolation
Anticoagulation	Yes, by CHA₂DS₂-VASc	Yes: TEE or anticoagulate before (> 48 h) and after

Table E: Cardioversion vs Defibrillation

	Synchronised cardioversion	Defibrillation (unsynchronised)
Timing	Delivered on the R wave (avoids R-on-T)	Random
Rhythms	Unstable SVT, AF (150 J), flutter, VT with pulse (200 J)	VF, pulseless VT
Asystole	—	✗ Not shockable → CPR + adrenaline

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Unstable tachycardia → synchronised DC shock	VF / pulseless VT → unsynchronised defibrillation; asystole → CPR (not shockable)
Adenosine for narrow regular SVT	Not in WPW with AF , not in asthma <i>[added]</i> ; it reveals but doesn't convert flutter
Verapamil for SVT	Never in wide-complex tachycardia or WPW
AV-nodal blockers control AF rate	★ AF + WPW → contraindicated → procainamide/DC
Wide QRS = VT	Could be SVT + BBB/aberrancy . Treat as VT if in doubt
Propafenone for recent AF	Not if CHF / impaired LV (use amiodarone)
Cardiovert AF	> 48 h → TEE or anticoagulate first ; anticoagulate after
CHA ₂ DS ₂ -VASc decides OAC	Valvular AF and HCM → anticoagulate regardless
Female sex = 1 point	Female sex alone (score 1) → low risk; thresholds are ≥ 3 for women
Atropine for bradycardia	Often ineffective in infranodal block / transplant → pace
Sinus arrhythmia	Respiratory variation is normal
Antiarrhythmics treat arrhythmias	Class Ia, Ic, III prolong QT → can cause VT/VF (pro-arrhythmia)
PVCs need treatment	Most don't: only frequent and symptomatic (β-blocker)

5. Clinical Correlations

- **Young WPW patient, binge drinking, then a regular wide-complex tachycardia at 220, stable** → **antidromic AVRT / pre-excited tachycardia** → **IV procainamide** (opening case).
- **Young woman with sudden palpitations, neck pounding, HR 180, narrow regular** → **AVNRT** → vagal → adenosine → ablation.
- **Inferior STEMI with HR 38 and Mobitz I** → AV-nodal ischaemia (RCA) → atropine if symptomatic; usually transient.
- **Anterior MI with new Mobitz II / CHB** → infranodal → **pacing**.
- **Elderly patient with AF, HTN, DM, age 76** → $\text{CHA}_2\text{DS}_2\text{-VASc} = \text{H1} + \text{D1} + \text{A}_2 2 = 4$ → **anticoagulate**.
- **AF in a patient with rheumatic MS** → **valvular AF** → **warfarin** regardless of score.
- **COPD exacerbation with irregular narrow tachycardia and ≥ 3 P-wave shapes** → **MAT** → treat COPD/hypoxia; avoid theophylline.
- **Patient on azithromycin + ondansetron with low $\text{K}^+/\text{Mg}^{2+}$, now polymorphic twisting VT** → **TdP** → **IV MgSO_4** .
- **Post-MI patient with sustained monomorphic VT and low EF** → amiodarone + **ICD**.
- **Thyrotoxic patient with AF** → β -blocker (propranolol) + treat the thyroid.
- **Digoxin toxicity** ([added] nausea, yellow vision, bidirectional VT, atrial tachycardia with block) → digoxin Fab.
- **Collapse: VF on the monitor** → **CPR + unsynchronised defibrillation**.

6. Rapid Revision (Last-Day)

1. Brady **< 60**, tachy **> 100**. SVT = SA node/atria/AV node; VT = below the AV node (**wide QRS**).
2. Brady causes: athletes, **SSS**, IHD, $\uparrow$ vagal tone, **RCA/inferior MI, digoxin/ β -blocker/CCB**.
3. Tachy causes: structural disease, ectopic foci, **accessory pathways, LQTS/Brugada/CPVT**, electrolytes, **COPD/PE/thyrotoxicosis, digoxin toxicity/theophylline**.
4. NSR: 60–100, P before every QRS, **PR 0.12–0.20, QRS < 0.12**.
5. AV blocks: **1st** PR **> 0.2** · **Mobitz I** progressive PR → drop · **Mobitz II** constant PR → drop · **3rd** dissociation, escape ~45.
6. Unstable brady ($\downarrow$ BP, confusion, shock, chest pain, HF): **atropine 1 mg q3–5 min (max 3 mg)** → **pacing or dopamine/adrenaline** → treat the cause → **PPM**.
7. Instability: **chest pain, altered mental status, pulmonary oedema, hypotension, shock** → **DC cardioversion**.
8. Four groups: **narrow regular** (ST, SVT, flutter, AT) · **wide regular** (VT $\pm$ SVT + AP/BBB) · **narrow irregular** (AF, MAT) · **wide irregular** (AF + AP/BBB, VF).
9. Sinus tach: treat the cause; **β -blocker** if persistent symptoms.
10. **AVNRT** = dual pathways, **60–70% of PSVT**, 150–250, neck pounding, P hidden. **AVRT** = accessory pathway; orthodromic narrow, antidromic wide.
11. Acute SVT: **vagal** → **adenosine 6 mg rapid IV ($\rightarrow 12 \rightarrow 12$)** → **verapamil 5 mg IV** → metoprolol/digoxin/diltiazem. AVRT: avoid digoxin and verapamil.
12. Long-term SVT: **RF ablation (> 98%, < 1% complications)**; 2nd line flecainide/propafenone. **WPW** → **EP study + ablation**.
13. AF types: paroxysmal **≤ 7 d**, persistent **> 7 d**, long-standing **> 1 y**, permanent.
14. AF unstable → **DC 150 J. < 48 h: propafenone 600 mg PO** (no HF) or **amiodarone 300 mg/40 min $\rightarrow$ 1800 mg/24 h. > 48 h: TEE, anticoagulate before and after (warfarin INR 2–3)**. Maintenance **amiodarone 200 / sotalol 80 bid**; recurrence → **PV isolation**.
15. Rate control: **β -blocker/non-DHP CCB** → **digoxin** → **amiodarone** → **AV-node ablation + PPM**.

16. **CHA₂DS₂-VASc**: C1 H1 **A₂ 2** D1 **S₂ 2** V1 A1 Sc1. **OAC if ≥ 2 ♂ / ≥ 3 ♀, valvular AF, HCM**; consider at 1 ♂ / 2 ♀.
17. **HAS-BLED** for bleeding risk (≥ 3 high). Bleeding risk must always be weighed.
18. **VT** = ≥ 3 ventricular beats ≥ 100 /min; NSVT < 30 s, sustained ≥ 30 s. Unstable → **DC 200 J**; acute MI → **lidocaine**; IHD → **amiodarone + ICD**; normal heart → **ablation**.
19. **TdP**: long QT; stable → **IV MgSO₄**; unstable → **defibrillation + CPR**.
20. **AF + WPW: no AV-nodal blockers or vagal manoeuvres → DC or IV procainamide**.
21. Carotid massage: supine, neck extended and rotated, **5-10 s**, repeat after 1 min, then the other side. Valsalva: **10-15 s** against a closed glottis.
22. **Synchronised** = AF/SVT/VT with pulse. **Defibrillation** = VF/pulseless VT. **Asystole → CPR**.
23. Classes: **I Na (IA quinidine/procainamide/disopyramide; IB lidocaine/mexiletine; IC flecainide/propafenone) · II β · III K (amiodarone, sotalol) · IV Ca (verapamil, diltiazem)**; misc: adenosine, digoxin, Mg, K.
24. Pro-arrhythmia: **Ia, Ic, III → QT/VT/VF; II, IV → bradycardia**.

7. Exam-Style Questions

Q1 (lecture opening case). A 16-year-old with a WPW pattern on sinus ECG (short PR, slurred upstroke) now has a **regular wide monomorphic tachycardia at 220/min**. BP 112/84, alert, no chest pain. What is the next step? A. Synchronised cardioversion · B. MgSO₄ · C. Verapamil · D. Atenolol · E. **Procainamide** · F. Adenosine

Answer: E. IV procainamide. [reasoning added; key not on slides]

- He has **WPW** with a **stable** wide-complex tachycardia (pre-excited / antidromic AVRT).
- Slide 77: **avoid AV-nodal blockers and vagal manoeuvres; rhythm control with cardioversion or IV procainamide is the safest**. As he is **stable**, chemical cardioversion with procainamide comes first.
- Synchronised DC (A) is for **instability**.
- **Verapamil, atenolol and adenosine** (C, D, F) block the AV node and can precipitate **VF**.
- **MgSO₄** (B) is for torsades.

Q2. A 72-year-old woman with hypertension and type 2 diabetes has new non-valvular AF. What is her CHA₂DS₂-VASc score and the recommendation? A. 2; no anticoagulation · B. 3; consider · C. **4; long-term anticoagulation** · D. 4; aspirin only · E. 5; anticoagulate

Answer: C.

- Score: **H 1 + D 1 + age 65-74 = 1 + female 1 = 4**.
- Women $\geq 3 \rightarrow$ **long-term anticoagulation** (slide 66).

Q3. A 25-year-old woman has sudden palpitations with neck pounding. HR 190, regular, narrow QRS, BP 118/76. Vagal manoeuvres fail. What next? A. DC cardioversion · B. **Adenosine 6 mg rapid IV push** · C. Amiodarone 300 mg · D. Digoxin · E. Lidocaine

Answer: B.

- **AVNRT**: adenosine is **first choice if vagal manoeuvres fail** (slide 51).
- DC is only if **unstable**.

Q4. A patient with AF of **unknown duration**, stable, rate 110, is planned for cardioversion. What must be done first? A. Immediate DC shock · B. Oral propafenone now · C. **TEE to exclude LA thrombus or ≥ 3 weeks of anticoagulation, then cardioversion and anticoagulation after** · D. Digoxin then discharge · E. Aspirin only

Answer: C.

- Unknown duration = treat as **> 48 h** → **exclude LA thrombus by TEE** and anticoagulate before and after (slide 62).

Q5. A hospitalised patient on haloperidol and azithromycin with K^+ 3.0 develops polymorphic VT twisting around the baseline. BP 110/70, conscious. What is the best immediate treatment? A. Amiodarone · B. Procainamide · C. **IV magnesium sulfate** · D. Adenosine · E. Unsynchronised shock

Answer: C.

- **Torsades** from acquired long QT; the patient is stable → **IV $MgSO_4$** (slide 72).
- **Amiodarone and procainamide prolong QT** and worsen it.
- Unsynchronised shock is for the **unstable/pulseless** patient.
- *[added]* Also stop the offending drugs and correct K^+ .

Q6. A 3rd-degree AV block with a wide ventricular escape at 35/min, BP 78/40 and confusion does not respond to atropine 3 mg. What next? A. More atropine · B. **Transcutaneous pacing (or dopamine/adrenaline infusion)** · C. Adenosine · D. Amiodarone · E. Observe

Answer: B.

- Atropine has failed (max 3 mg reached) → **temporising pacing or dopamine/epinephrine**, then a permanent pacemaker (slide 23).

8. Exam Pitfalls & Traps

- **Synchronised vs unsynchronised:** pulse present → synchronised. VF/pulseless VT → defibrillation. **Asystole is never shocked.**
- **Energy numbers on the slides: AF DC 150 J; VT DC 200 J.** *[added]* Device-specific (biphasic); quote the slide numbers.
- **Slide 67 HAS-BLED caption:** the 0 / 1-2 / ≥ 3 bands sit under a **HAS-BLED (bleeding)** table, on a slide titled "thromboembolic". They refer to **bleeding risk**. Stroke risk uses CHA_2DS_2-VASc thresholds (slide 66).
- **CHA_2DS_2-VASc :** age $\geq 75 = 2$ and stroke = **2**. Female sex alone doesn't trigger OAC. Thresholds: **≥ 2 men / ≥ 3 women.**
- **Valvular AF / HCM** → anticoagulate **regardless of score**. *[added]* Warfarin (not DOAC) for mechanical valves or moderate-severe MS.
- **AVRT acute treatment:** the slide says avoid **long-acting** AV-nodal blockers (digoxin, verapamil). In **pre-excited AF**, **all** AV-nodal blockers + vagal manoeuvres are contraindicated (slides 74, 77).
- **Orthodromic = narrow; antidromic = wide** with delta waves.
- **Slide 84 vs slide 77 on pre-excited AF:** the older algorithm (84) says "consider amiodarone". The lecture's WPW slide (77) says **procainamide or cardioversion**, which is the current answer. *[added]* IV amiodarone has AV-nodal effects and is now discouraged.
- **Verapamil in a wide-complex tachycardia** can cause collapse. Treat as VT unless proven SVT with aberrancy.

- **Slide 53 PSVT rate 140-280** vs **slide 49 AVNRT 150-250**: different sources. Either range is acceptable; know "~150-250".
- **Slide 51** writes "Isoptin": that is **verapamil**. The "precautions ??" for carotid massage = **bruit, recent stroke/TIA, elderly with carotid disease** *[added]*.
- **Propafenone** (class IC) is contraindicated in **HF / structural heart disease / IHD**; use amiodarone.
- **Atrial flutter** with 2:1 block gives a regular **~150** rate. Any regular narrow tachycardia at exactly 150 → think flutter.
- **Respiratory sinus arrhythmia** is normal. Don't treat it.
- **Mobitz I vs II**: the PR **changes** in I and is **constant** in II. Mobitz II needs a pacemaker.